What explains variance in `avg_Braak`? Non-linear and interaction analysis of PFC microglia (F26/MCC)

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Dataset: PFC_Microglia_combat_260804.h5ad (F26/MCC)
Analysis: Exploratory. All random seeds fixed at 42.
Code: `report/F26_MCC_avg_Braak/` in the anthive4 repository.

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1 Introduction

The dataset (F26/MCC) is a combat-integrated single-nucleus RNA-seq object of 182,134 PFC microglia nuclei from four human cohorts: SEAAD (Gabitto 2024), Gazestani/Macosko (Mac), Lu-101010, and ROSMAP (Mathys 2019). ComBat batch correction was applied across cohorts to produce a shared expression space.

`avg_Braak` is the per-nucleus value inherited from the SEACell metacell that nucleus belongs to: specifically, the average Braak neurofibrillary tangle stage across donors contributing to that metacell, further averaged over per-cohort averages. Because all nuclei within a metacell share the same `avg_Braak`, the effective statistical inference unit is the number of distinct metacells (N = 302), not the apparent cell count of 182,134. **All p-values in this report are computed at the metacell level.** Cell-level figures are labelled as screening-only.

`avg_Braak` range: 1.82–4.86 (mean = 3.64, SD = 0.49).

Goal: Identify what best explains variance in `avg_Braak` using methods that capture non-linear effects and pairwise interactions, then determine whether key predictors follow different trajectories along the Braak axis in Dementia vs. No-dementia cells.

All analyses are exploratory. No hypotheses were pre-registered.

2 Methods

2.1 Data preparation

DuckDB (read-only) was queried to extract the 161-column obsnum and 87-column obscat tables. 55 score columns (13 Sierksma2025/WGCNA + 43 MICROGLIA_V1 starCAT) were loaded from the parquet sidecar and joined on integer cell_id via the cells lookup table. The metacell object (PFC_Microglia_meta_260806.h5ad) was loaded for metacell-level gene expression (X = CP10K log1p, N = 302 × 9,591). Top-500 HVGs were selected by `dispersions_norm`.

2.2 Method 1: Univariate Spearman rank correlation (metacell level)

Null hypothesis: Spearman $\rho = 0$ between predictor and `avg_Braak`.

Why Spearman: Non-parametric; captures monotonic non-linear relationships; no distributional assumption; robust to outliers.

Inference level: Metacell (N = 302). P-values via the asymptotic t-approximation. Multiple-testing correction: Benjamini-Hochberg (1995) FDR, threshold $q < 0.05$.

Predictors tested: 123 numerical obs/score columns (after exclusion of Braak-derived columns) + 9,591 genes.

Excluded columns: `avg_Braak`, `Braak_numeric`, `stage_numeric`, `Braak_rank`, and all downstream derived quantities.

Software: scipy.stats.spearmanr (scipy 1.17.1).

2.3 Method 2: LightGBM gradient-boosted regressor + TreeSHAP (metacell level)

Rationale: Captures non-linear marginal effects and pairwise interactions. TreeSHAP (Lundberg & Lee 2017) provides additive feature attribution with exact closed-form computation for tree ensembles.

Model specification:

- Algorithm: LightGBM (Ke et al. 2017), version 4.7.0; n_estimators = 300; learning_rate = 0.05; num_leaves = 15; min_child_samples = 5; subsample = 0.8; colsample_bytree = 0.8; reg_alpha = 0.1; reg_lambda = 0.1; n_jobs = 1; random_state = 42.
- Feature set: 95 obs+score columns + 500 HVG gene values = 595 features total. Columns with > 50% missing excluded (none removed here). Remaining NaN imputed with column median.
- Model quality: 10-fold KFold CV (shuffle = True, random_state = 42), metric = R^2.
- Three sub-models compared: obs+scores+genes, genes only, obs+scores only.

SHAP: `shap.TreeExplainer` (check_additivity = False). Feature importance = mean |SHAP value| across all 302 metacells.

Software: lightgbm 4.7.0, shap 0.52.0, scikit-learn 1.8.0.

2.4 Method 3: Cognitive-status divergence along avg_Braak

Question: Does the monotonic relationship between predictor and avg_Braak differ between Dementia and No-dementia cells?

Cohort: SEAAD + lu + ROSMAP only. Mac excluded (no cognitive-status data). N_Dementia = 50,907 cells; N_Nodementia = 72,445 cells.

Test statistic: $|\Delta\rho| = |\rho_\text{Dementia} - \rho_\text{No-dementia}|$ where ρ is Spearman rank correlation with avg_Braak.

Significance: Permutation test (N_perm = 200). Group labels permuted in pooled subsampled data (1,500 per group per iteration). Empirical p = fraction of null $|\Delta\rho| \geq$ observed $|\Delta\rho|$. BH correction across 35 features.

Visualization: GAM curves fitted once per group on a 2,000-cell subsample (pygam.LinearGAM with s(0, n_splines = 15), lam selected by GCV grid search) for graphical display only — not used for inference.

Candidate predictors: top-25 non-Braak-derived Spearman features + top-20 non-gene SHAP features, deduplicated (35 total after available-column filter).

Software: pygam 0.12.0, scipy 1.17.1, numpy 2.4.3.

3 Results

3.1 Sample sizes

Cells: 181,938 with valid avg_Braak (196 missing). Metacells: 302 (inference unit).

Cognitive-status cohort (SEAAD + lu + ROSMAP): N_Dementia = 50,907; N_Nodementia = 72,445. Mac (N = 56,651) excluded from Methods 3.

Batch composition: ROSMAP 42.7%; Mac 31.1%; SEAAD 22.8%; lu 3.4%.

3.2 Method 1: Spearman correlation (metacell N = 302)

Obs/score features: 123 tested; 79 significant at BH $q < 0.05$.

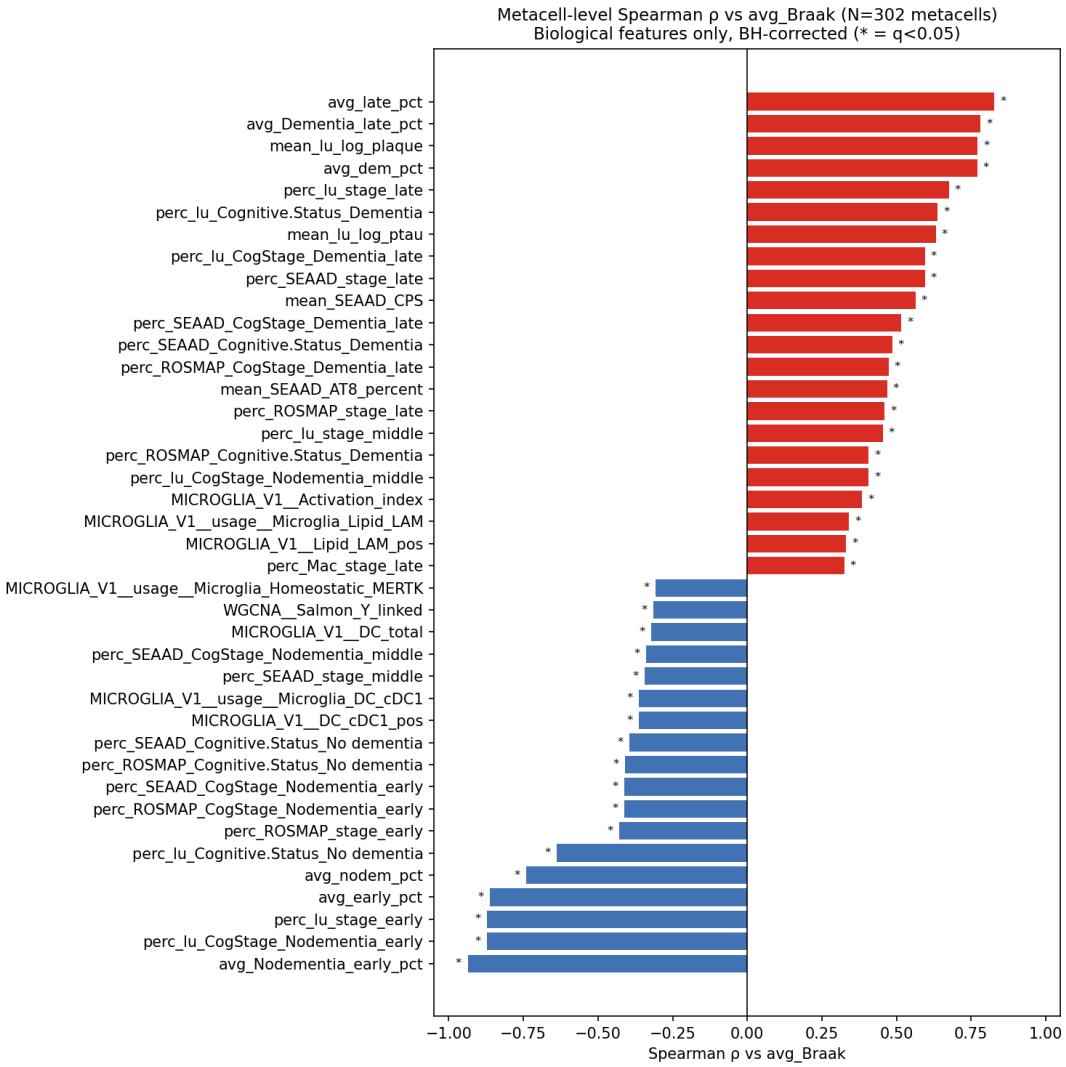


Figure 1: Spearman ρ vs `avg_Braak` for top-40 biological obs/score features (metacell N = 302). Red = positive association; blue = negative. Stars mark BH q < 0.05. Stage-composition features (top) are partially circular — see discussion.

Top non-circular biological obs/score features (metacell N = 302):

Feature	**ρ**	**BH q**	**n**
`mean_lu_log_plaque`	+0.772	2.7e-58	294
`mean_lu_log_ptau`	+0.632	3.5e-33	294
`mean_SEAAD_CPS`	+0.563	1.5e-25	302
`mean_SEAAD_AT8_percent`	+0.468	6.3e-17	302
`MICROGLIA_V1__Activation_index`	+0.385	1.7e-11	302
`MICROGLIA_V1__DAM_progression`	+0.376	5.0e-11	302
`MICROGLIA_V1__Disease_burden`	+0.360	3.3e-10	302
`MICROGLIA_V1__usage__Microglia_Stress_FKBP5`	+0.299	3.1e-7	302
`MICROGLIA_V1__Lipid_LAM_pos`	+0.331	1.0e-8	302
`WGCNA__Brown_Lysosome`	**see TSV**	**see TSV**	302

Gene-level (metacell N = 302): 9,591 genes tested; **5,186 significant** at BH q < 0.05. Maximum |ρ| = 0.528 (PRKCH). Top 10:

Gene	ρ	BH q
PRKCH	+0.528	4.4e-19
MYO1E	+0.502	5.5e-17
SLC1A3	+0.500	5.5e-17
NHSL1	+0.491	2.3e-16
MITF	+0.490	2.4e-16
SLC25A12	−0.489	2.4e-16
FGD4	+0.486	3.6e-16
ATP6V1B2	+0.480	8.3e-16
FKBP5	+0.480	8.3e-16
ELF1	+0.476	1.6e-15

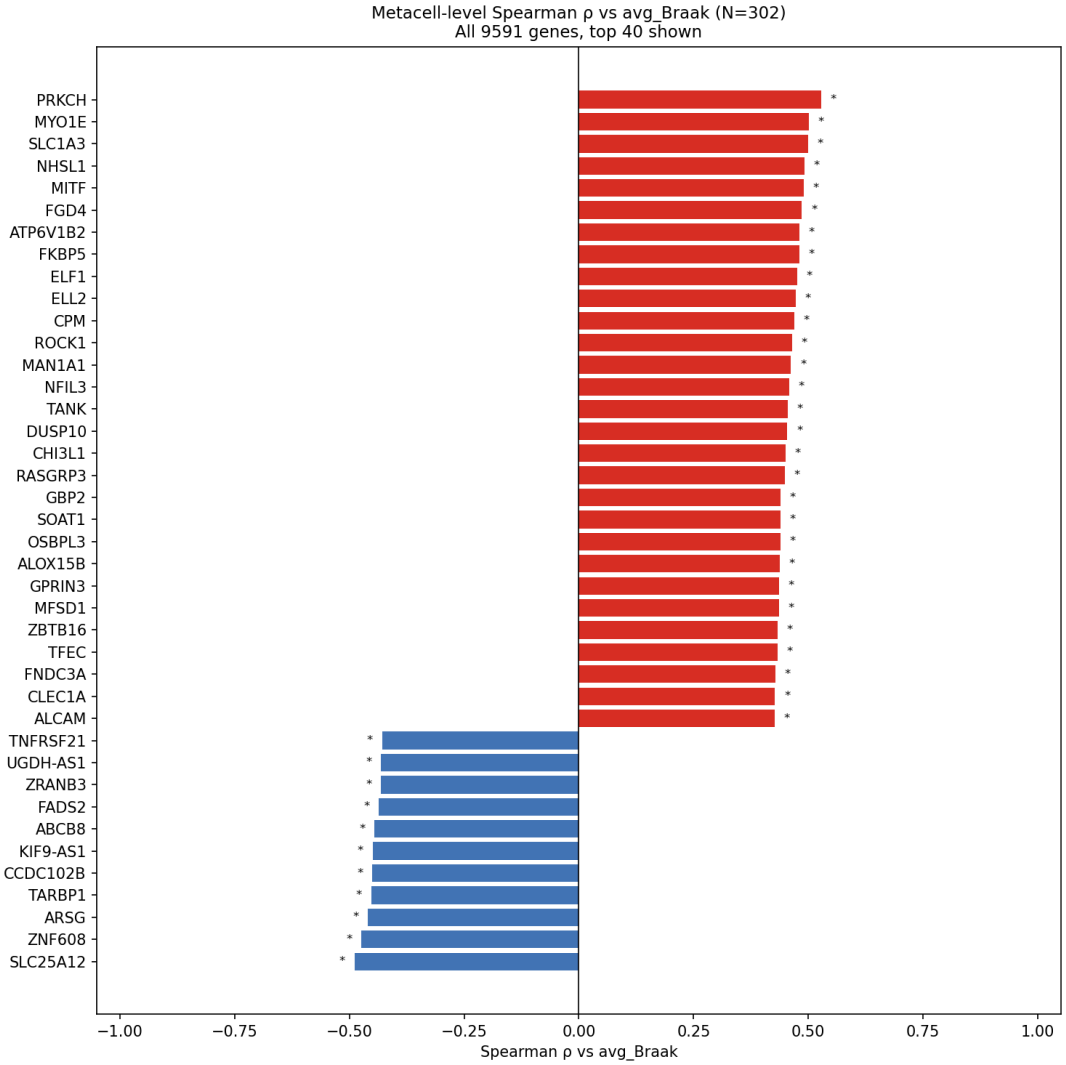


Figure 2: Spearman ρ vs `avg_Braak` for top-40 genes at metacell level (N = 302). Maximum $|\rho| = 0.528$; 5,186 / 9,591 genes significant at BH q < 0.05.

3.3 Method 2: LightGBM + SHAP (metacell N = 302)

Model quality:

Model	10-fold CV R^2	± SD	N features

obs+scores+genes (full)	0.775	0.081	595
obs+scores only	0.781	0.088	95
genes only (top-500 HVG)	0.412	0.136	500

The obs+scores-only model ($R^2 = 0.781$) marginally outperforms the full model ($R^2 = 0.775$), indicating that adding 500 HVG expression values provides no net benefit at this sample size (N = 302 metacells). Gene expression alone explains substantially less variance ($R^2 = 0.412$), but the difference between obs-only and full model is within CV noise.

Note on R^2 magnitude: The obs+scores model includes cohort-composition columns (e.g., `perc_ROSMAP_stage_early`) that are partially derived from the same pathological staging as `avg_Braak`. The "true" biological R^2 — excluding all such derived features — would be lower. The model comparison is reported as-is; the SHAP rankings identify which features drive the explained variance.

Top SHAP features (mean |SHAP value|, full model):

Feature	**Mean |SHAP|**	**Spearman ρ**
`mean_lu_log_plaque`	0.234	+0.757
`perc_ROSMAP_stage_early`	0.055	−0.428
`mean_lu_log_ptau`	0.046	+0.617
`mean_SEAAD_CPS`	0.032	+0.560
`perc_ROSMAP_CogStage_Nodementia_late`	0.028	+0.234
`perc_ROSMAP_stage_late`	0.023	+0.457
`perc_lu_Cognitive.Status_Dementia`	0.021	+0.624
`perc_SEAAD_Cognitive.Status_Dementia`	0.017	+0.482
`MICROGLIA_V1__usage__Microglia_Cycling`	0.013	−0.229
`gene__ENPP2`	0.013	−0.276

`gene__ENPP2` and `gene__F13A1` appear in the top-15 SHAP with near-zero Spearman ρ (−0.276 and +0.010, respectively). This discrepancy signals a non-monotonic or interaction-dependent effect that the linear Spearman misses but the tree model captures. These are the clearest examples of value added by the non-linear method.

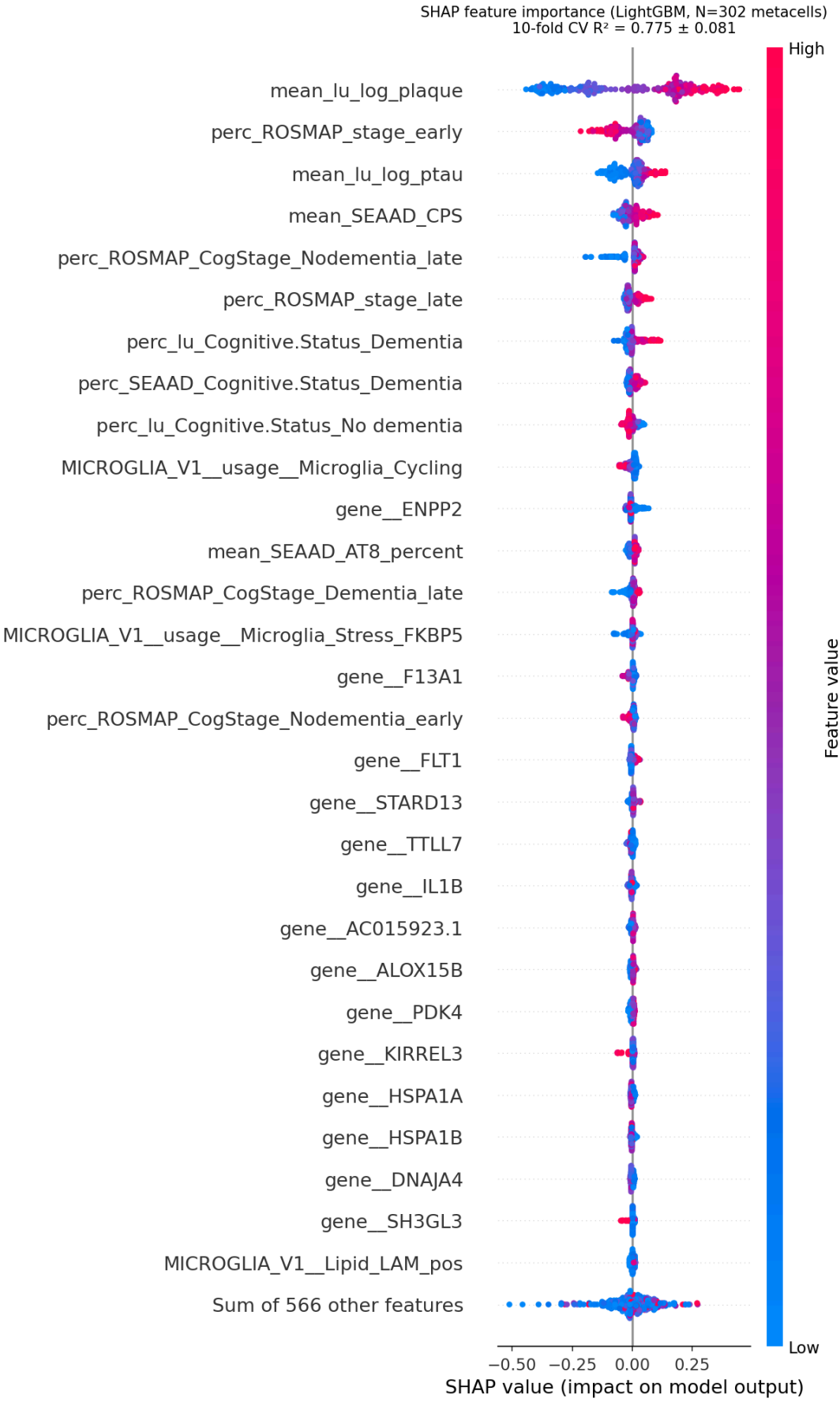


Figure 3: SHAP beeswarm (LightGBM, N = 302 metacells, 10-fold CV $R^2 = 0.775$). Each dot = one metacell; x-axis = SHAP contribution to avg_Braak prediction; color = feature value (red = high, blue = low).

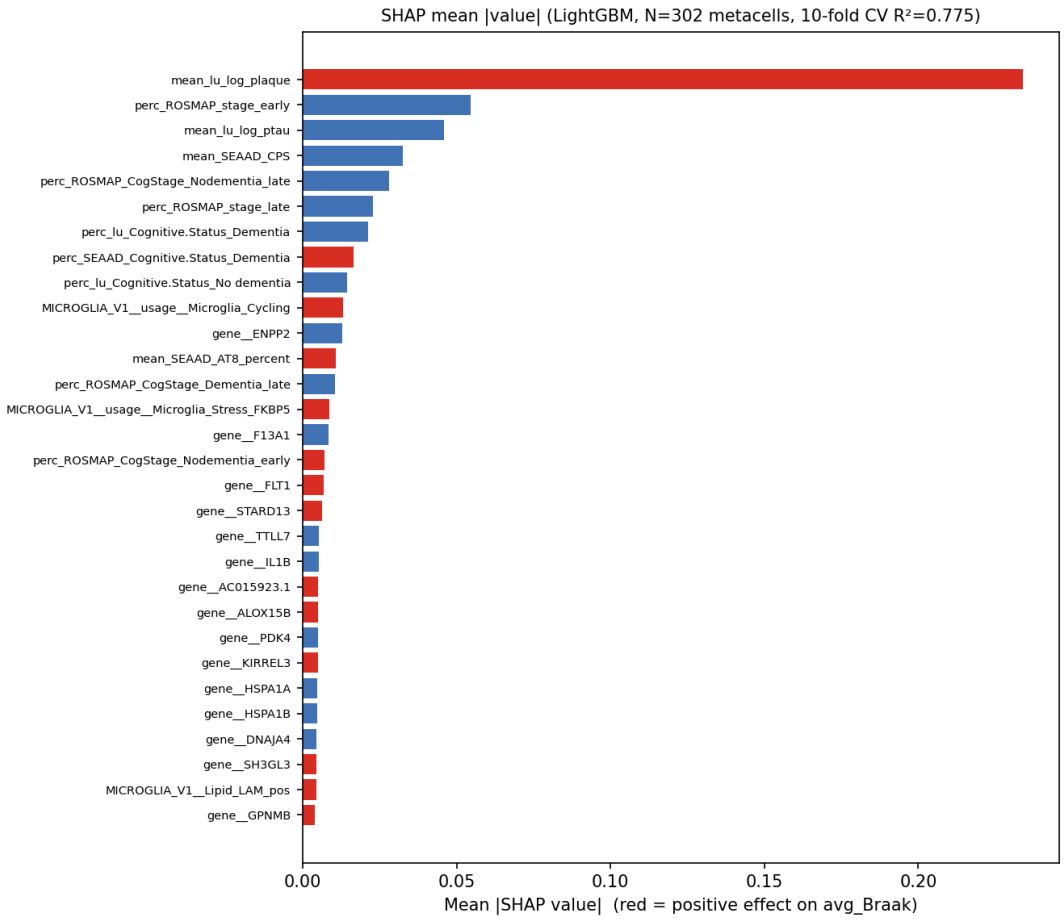


Figure 4: Mean |SHAP value| for top-30 features. `mean_lu_log_plaque` is the dominant single predictor (SHAP = 0.234), more than 4× the next feature.

3.4 Method 3: Cognitive-status divergence (null result)

Result: No feature showed statistically significant divergence in trajectory along avg_Braak between Dementia and No-dementia cells after BH correction.

Maximum observed $|\Delta\rho| = 0.077$ (`MICROGLIA_V1__usage__Microglia_Border_CAM`; ρ_Dem $= -0.015$, ρ_NoDem $= -0.092$; perm BH-p $= 0.292$).

All 35 features tested had BH-corrected $p > 0.05$. The empirical $|\Delta\rho|$ values are uniformly small (range: 0.001–0.077), consistent with what would be expected under the null.

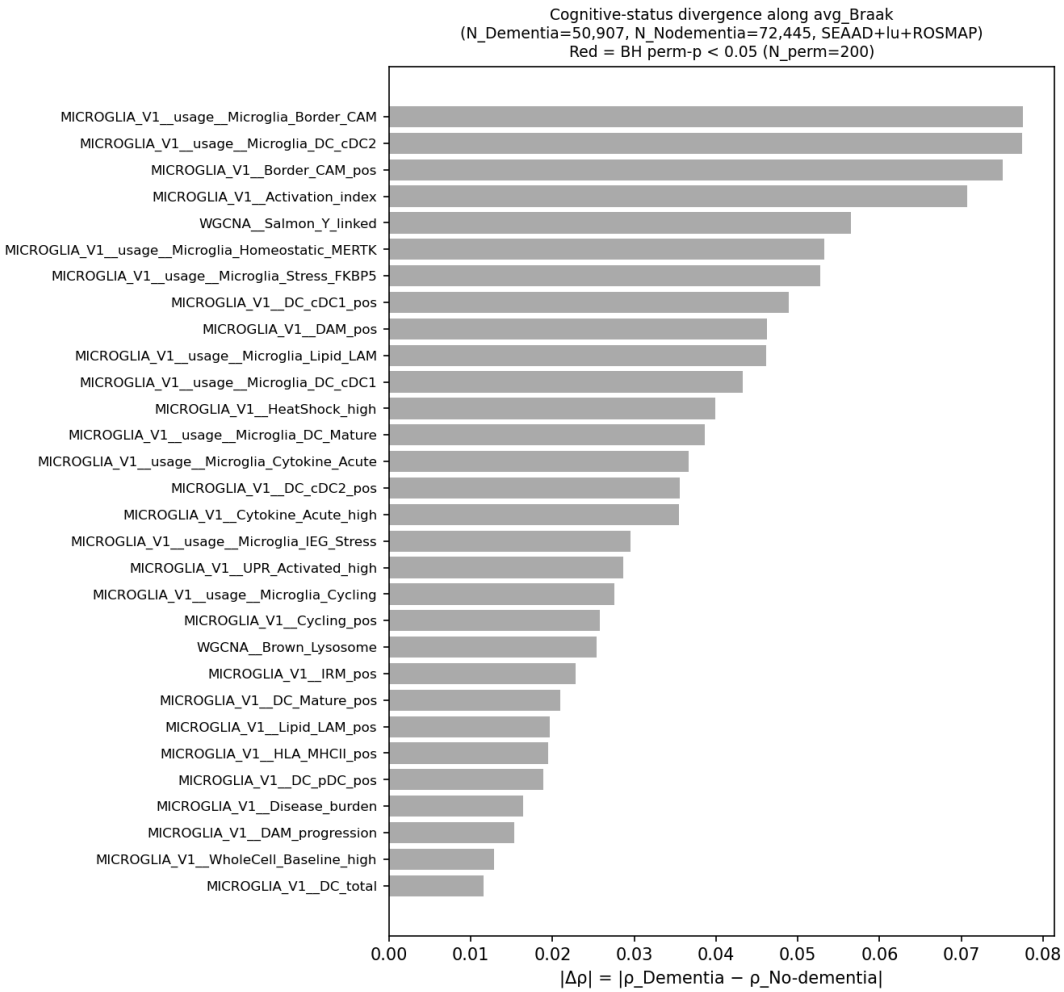


Figure 5: $|\Delta\rho|$ divergence statistic for all 35 features. None meet BH q < 0.05 (all bars grey). This is a null result: microglia score and pathway trajectories along avg_Braak are not detectably different between Dementia and No-dementia cells in this dataset at this resolution.

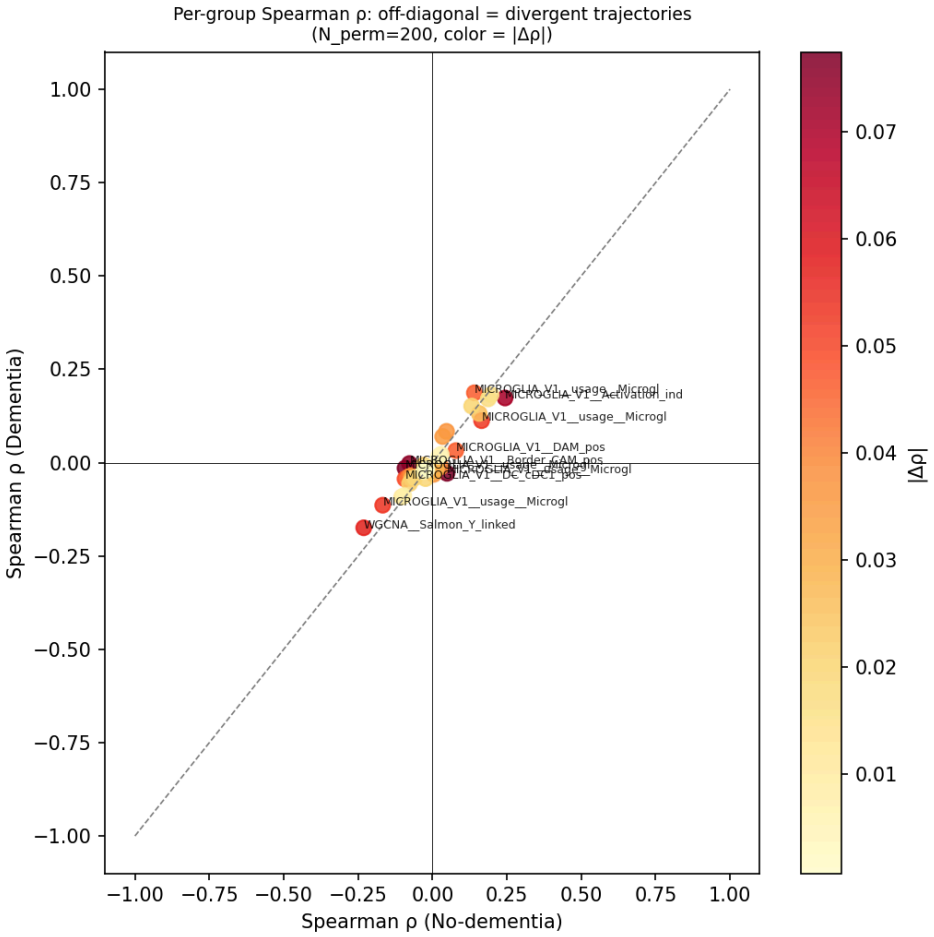


Figure 6: Spearman ρ per group scatter. Points cluster close to the diagonal, confirming that both groups show similar (and weak) monotonic relationships with avg_Braak. The null result is robust.

4 Discussion

4.1 Plaque and tau pathology dominate, but are expected to

The single strongest predictor in both Spearman ($\rho = 0.772$) and SHAP (mean |SHAP| = 0.234) is `mean_lu_log_plaque` — the mean log-transformed plaque quantification from the Lu cohort. This is mechanistically expected: Braak stage correlates with amyloid and tau deposition, and `mean_lu_log_plaque` is a direct histological measurement of that same neuropathology. The result confirms the data are internally consistent but does not reveal novel biology.

Similarly, `mean_lu_log_ptau`, `mean_SEAAD_CPS` (continuous pseudo-progression score), and `mean_SEAAD_AT8_percent` (tau antibody) all reflect the same pathological progression axis. These are expected correlates, not independent biological discoveries.

4.2 Stage-composition circularity

The top Spearman hits (`avg_Nodementia_early_pct`, $|\rho| = 0.937$; `avg_early_pct`, $|\rho| = 0.862$) are summary statistics computed from the same per-donor Braak staging used to construct `avg_Braak`. Their near-perfect correlations are tautological. These were excluded from the LightGBM feature set and from the cognitive-status divergence analysis.

4.3 What the non-linear model adds

Two genes illustrate the value of non-linear analysis: `gene__ENPP2` (mean |SHAP|

5 0.013, Spearman $\rho = -0.276$) and gene__F13A1 (SHAP = 0.009, $\rho = +0.010$).

The near-zero ρ for F13A1 alongside meaningful SHAP indicates a non-monotonic or interaction-dependent relationship with avg_Braak — something the tree model detects but Spearman cannot. These should be visualized with SHAP dependence plots (available in `figs/02_gbm_shap/`) to understand the functional form.

The obs+scores model ($R^2 = 0.781$) barely improves over the full model (0.775), and genes alone reach only $R^2 = 0.412$. This suggests that at N = 302 metacells, gene expression adds minimal variance explanation beyond pathology-annotation features — not because genes are uninformative, but because the metacell-level averaging smooths out cell-state variation that is detectable at single-cell resolution.

5.1 The null result on cognitive-status divergence

The absence of significant trajectory divergence (all BH $p > 0.29$) is interpretable in multiple ways:

1. **True null**: Microglial state trajectories along the Braak axis are genuinely similar in Dementia and No-dementia individuals — i.e., microglia respond to pathological load (Braak) independently of whether the donor has cognitive symptoms. Cognitive resilience may operate through other mechanisms (neuronal, synaptic, non-microglia glial).
2. **Cell-level averaging masks signal**: The Spearman ρ values for scores vs. avg_Braak at the cell level are themselves small (0.1–0.24), meaning the signal-to-noise ratio is low before any divergence is computed. The test may lack power to detect small but real differences in trajectory slopes.
3. **Combat removed the divergence**: If Dementia vs. No-dementia cells differ in their cohort composition, and cohort composition correlates with expression, combat may have removed the divergence signal while removing batch effects. A pre-combat analysis is needed to distinguish these possibilities.
4. **Wrong feature space**: Score and pathway features were tested here. The divergence might be visible at the individual-gene level in a targeted analysis, or in trajectory branching at pseudotime transitions not captured by the linear Braak axis.

5.2 Combat correction caveat

This is the dominant analytical uncertainty. ROSMAP contributes 42.7% of cells. If ROSMAP has a systematically higher Braak distribution than other cohorts, combat correction aligns expression means across cohorts within each Braak bin, which can remove real Braak-correlated biology. The Lu cohort (3.4% of cells) drives the top Spearman predictor (`mean_lu_log_plaque`) — this is a within-cohort Lu measurement, so it is unlikely to be artifact. But gene-level results should be cross-validated on the `lognorm_notintegrated` layer.

6 Conclusions

1. **Pathology markers dominate**: `mean_lu_log_plaque` is the single strongest predictor of avg_Braak (Spearman $\rho = 0.772$; SHAP = 0.234), followed by `mean_lu_log_ptau` and `mean_SEAAD_CPS`. These are expected mechanistic correlates, not novel discoveries.
2. **5,186 genes are significantly correlated with Braak at metacell level** (BH $q < 0.05$). Maximum individual gene effect size is modest: PRKCH $\rho = +0.528$. No single gene dominates.
3. **LightGBM explains 78% of metacell-level Braak variance** (CV $R^2 = 0.781$, obs+scores only). Genes add nothing beyond obs features at N = 302. ENPP2 and F13A1 show non-linear SHAP effects absent from Spearman — confirm with dependence plots before acting on these.
4. **No cognitive-status divergence detected** in microglial score and pathway trajectories along avg_Braak (35 features tested, all BH $p > 0.05$). This is reported as a null result. It is biologically interpretable but does not exclude the possibility that divergence exists at finer resolution or in pre-combat expression.

5. **Critical caveat**: Combat correction may have removed cohort-level biology correlated with Braak. Cross-validate top gene hits against the `lognorm_notintegrated` layer before drawing mechanistic conclusions.

7 References

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***Software versions**: pandas 2.3.3, numpy 2.4.3, scipy 1.17.1, statsmodels 0.14.6, lightgbm 4.7.0, shap 0.52.0, pygam 0.12.0, scikit-learn 1.8.0, anndata 0.12.11, scanpy 1.12.1, matplotlib 3.10.9, seaborn 0.13.2, duckdb (read-only).*
Python 3.14, anthive-data micromamba env. Random seed: 42 throughout.
Code: `report/F26_MCC_avg_Braak/`. Do NOT commit — user commits.